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Frequency Spectra and Wave Height Distributions Based on In-Situ Measurements

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Introduction

This thesis continues the long-term investigation of sea surface wave activity near Sakhalin Island, based on bottom pressure variation stations that have been operated by the Specialized Design Bureau of the Sakhalin Branch of the Far Eastern Branch of the Russian Academy of Sciences for the past fifteen years. In [2], a probability distribution of wave heights was constructed from measurements collected between 2012 and 2015, showing a strong agreement with the theoretical distribution proposed by Glukhovsky [1], [4], [14]. The Glukhovsky distribution provides corrections to the Rayleigh model by considering finite depth effects, expressed through a nonlinear distribution parameter.

In [5], it was discussed that due to variability in the wave period and the presence of tides, measurements taken at a single location in fact corresponded to a wide range of dimensionless depths, from relatively small to moderately large values ($0.3 \leq kh \leq 2$, where h is the local depth and k is the characteristic wavenumber). Such variability results in different conditions for wave propagation physics, particularly with respect to dispersion and nonlinearity, which may manifest as heterogeneity in the statistical properties of waves. The dependence of the probability distribution of wave heights on dimensionless nonlinear parameters was analyzed in [5], where it was shown that the distribution evolves systematically with increasing root-mean-square surface displacement σ , normalized by the local depth, $a = \sigma/h$. As the nonlinearity parameter a increases, the probability of high waves decreases significantly. Thus, in shallow water, nonlinearity can be seen to reduce the likelihood of extreme waves occurring.

Such a nonlinearity parameter is natural for shallow-water conditions, where two characteristic spatial scales, wave height and depth, form a dimensionless quantity. It approximately corresponds to the nonlinear parameter incorporated in the Glukhovsky distribution. The reduction in the fraction of high waves in shallow water with increasing nonlinearity is qualitatively opposite to the case of deep water, $kh \gg 1$, where both theoretical and experimental studies have demonstrated an increase in the probability of high waves (the so-called rogue waves) with increasing mean wave steepness $\varepsilon = k\sigma$, which characterizes nonlinearity in this regime [11], [17], [25]. The generally accepted mechanism responsible for wave amplification in deep water is nonlinear self-modulation [6]. At the same time, for shallow-water conditions, numerical modeling of the weakly nonlinear Korteweg–de Vries equation in [21] revealed regimes in which both the fourth statistical moment (excess and kurtosis) and the fraction of large-amplitude waves increase at significant values of the Ursell parameter, which characterizes the relative roles of wave nonlinearity and dispersion.

The considerable variability observed in field data evidently concerns not only the dominant wave period and wave intensity, which are usually regarded as the two most important characteristics of sea

states, but also the spectral distributions of wave energy. In particular, the experimental study [26] demonstrated a decrease in the probability of high waves in deep water as the spectral bandwidth increases. The ambiguity in defining spectral width and the significant role of spectral shape with respect to the probabilistic properties of high waves were discussed, among others, in [24] and [26]. In oceanography, the empirical Pierson–Moskowitz spectrum and its later generalization, JONSWAP [4], [14], are widely used. In practice, these spectra are employed to parameterize wave spectra both in deep and shallow water. The Pierson–Moskowitz spectrum is governed by two parameters of sea state, the significant wave height, and the peak period, while the JONSWAP spectrum includes, in addition, the so-called peak enhancement factor, which determines the width of the pronounced spectral peak. In natural conditions, multiple wave systems may coexist, differing in periods and/or intensities, as well as in propagation directions. From measurements at a single point by one sensor, only a frequency spectrum can be constructed, in which waves propagating along different azimuthal directions are mixed and cannot be reliably separated. The principal feasibility of reconstructing directional spatial spectra from wave records using a distributed synchronized array of bottom pressure sensors was demonstrated in our work [3]. For a two-day record, the reconstructed angular spectrum exhibited a well-defined dominant direction, and its shape was consistent with the widely used $\cos^2$ parameterization [14], with a width parameter in the range of $50 - 90^\circ$ (out of the maximum possible 180°). However, all other wave measurements near Sakhalin were conducted with single instruments, and therefore information on directional spectral distributions is not available for those records.

In this study, we examine the shapes of frequency spectra derived from measurements near Sakhalin Island, considered as the next natural characteristic of sea states after the representative period and wave height. The simplest parameter for a more detailed description of the frequency spectrum is its bandwidth. As discussed above, this parameter is of particular interest because of its potential influence on the probability distribution of waves, which we address separately. In this context, a fragment of theory concerning the dependence of wave amplitude distribution curves on spectral width will also be presented. For the present analysis, we use the data set previously examined in [5] (measurements near Cape Svobodny from 2012 to 2015, obtained with single pressure sensors at a sampling rate of 1 Hz), supplemented by measurements from 2020 to 2022 collected at approximately the same location with a sensor sampling rate of 8 Hz.

The data preparation procedure is described in [29]. Sea surface displacement was estimated from bottom pressure measurements within the framework of hydrostatic theory. The original time series were filtered to remove long-period oscillations (longer than 10 minutes), which are associated with tides, atmospheric fronts, and similar processes, using a spectral filter. For statistical analysis, the records were divided into 20-minute segments, assumed to represent statistically homogeneous data. Each segment was

processed independently, and mean values were determined for parameters such as mean wave frequency, root-mean-square surface displacement from the undisturbed level, and others. In this study, the wave spectrum was calculated using the Fourier transform without window averaging [14], unless otherwise specified. Individual wave heights were determined using the up-crossing method, based on intersections of the undisturbed water level in the data sequence.

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1. EVOLUTION OF FREQUENCY SPECTRA AND THEIR CHARACTERISTIC PROFILES

Time–frequency wave spectra provide insight into the ongoing processes of changing wave conditions. Several examples of “effect-rich” time–frequency spectra, constructed from 10-minute segments with a 1-minute step, are shown in Fig. 1. The diagrams clearly reveal instances of multiple wave systems coexisting simultaneously (manifested as distinct spectral maxima). Particularly in the lower panel throughout the entire record exceeding one month. In the middle panel, for times $t > 100$ h, the emergence and development of a wave system with a decreasing dominant frequency can be observed. In the lower panel, for times $t > 200$ h, the opposite situation is apparent, where the dominant frequency of the system increases from a value below 1 rad/s, corresponding to waves arriving from a distant source, initially longer and faster waves.

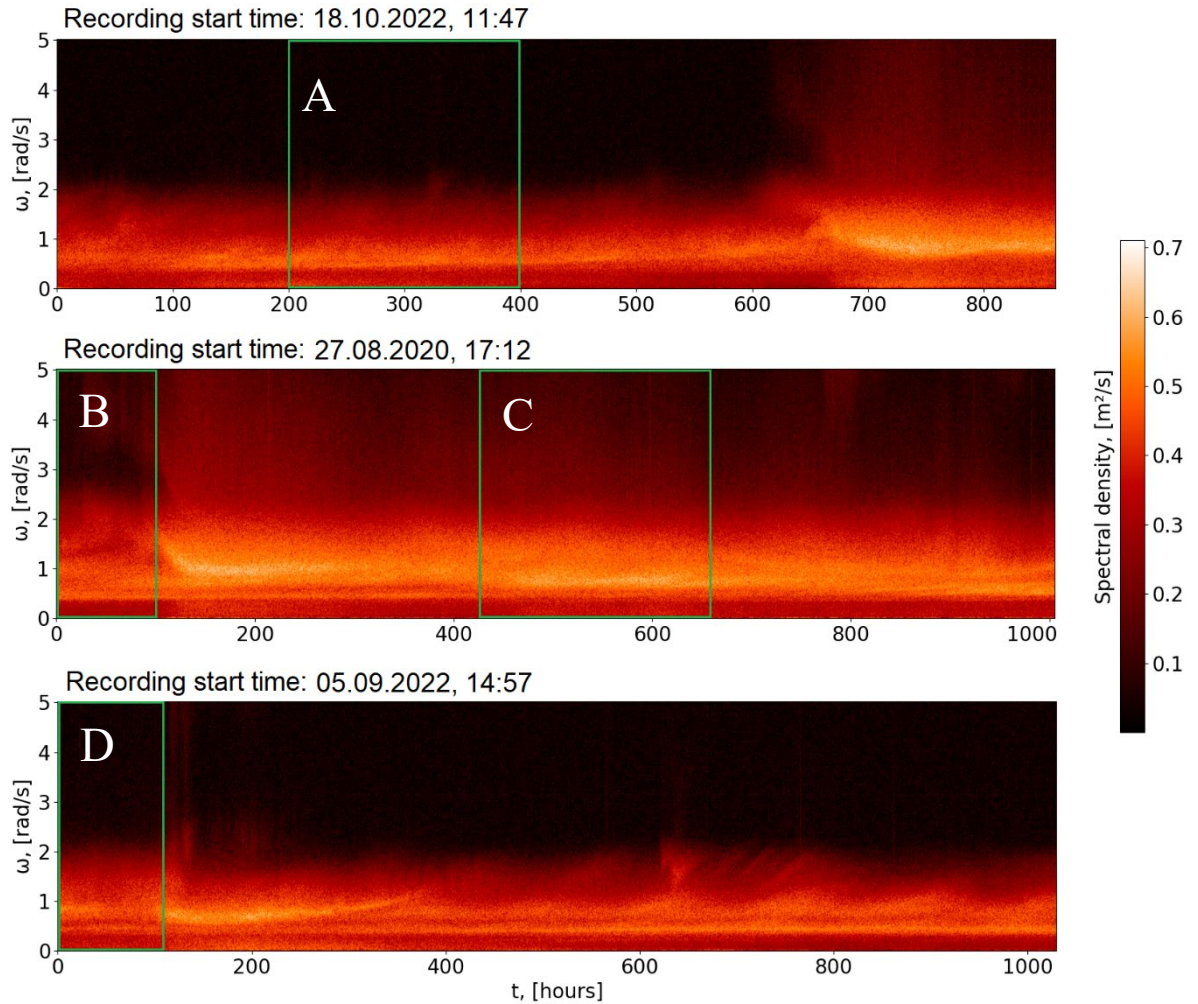


Fig. 1. Examples of the evolution of energy frequency spectra $S(\omega)$. The color represents the quantity $\log(|S|/\max(|S|))$ normalized by its maximum, while the color scale indicates values of $|S|/\max(|S|)$.

Fig. 2 shows the frequency spectra for time intervals A, B, C, and D, marked by rectangles in Fig. 1, which correspond to approximately stationary conditions (according to the spectral portraits) lasting from 100 to 200 hours [31]. Among these, one can observe intense wave activity with a single pronounced spectral peak corresponding to a period of about 10 s (example C); two-peaked spectra (examples A and B) with large and slight differences between the peak wave periods. In example B, a third, even lower-frequency peak can also be visually identified.

The shape of the single-peaked spectrum in case C corresponds satisfactorily to the JONSWAP model spectrum [10]:

$$S(\omega) = \alpha \left(\frac{\omega}{\omega_p} \right)^{-5} \exp \left[-\frac{5}{4} \left(\frac{\omega}{\omega_p} \right)^4 \right] \gamma \exp \left[-\frac{1}{2\beta^2} \left(\frac{\omega - \omega_p}{\omega_p} \right)^2 \right], \quad \beta(\omega) = \begin{cases} 0.07, & \omega < \omega_p \\ 0.09 & \omega > \omega_p \end{cases}, \quad (1)$$

where the parameter α is related to wave intensity, ω_p is the spectral peak frequency, and γ is the peak enhancement factor. In Fig. 2C, lines show spectrum (1) for several values of the peak enhancement factor, demonstrating reasonable agreement with the measured data. Visually, the best fit is achieved for $\gamma = 1.33$.

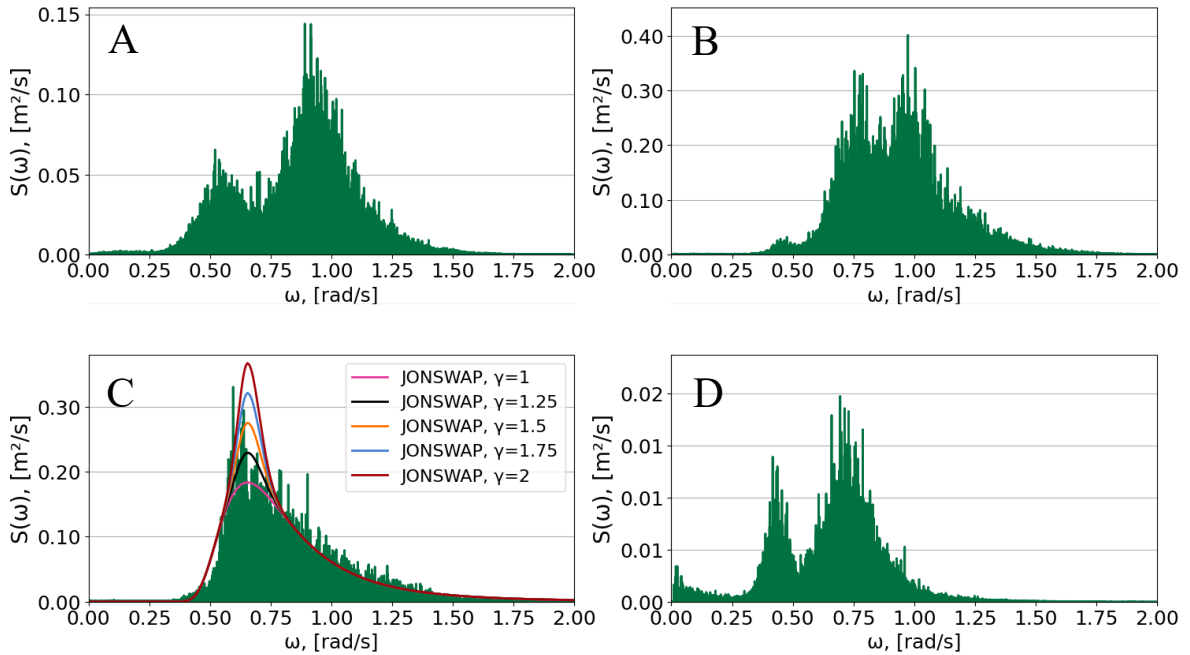


Fig. 2. Frequency spectra for the intervals highlighted by rectangles in Fig. 1. In panel C, a JONSWAP model spectrum is additionally plotted for several values of the peak enhancement factor.

In Fig. 2D, in addition to the two peaks corresponding to periods of 8 s and 14 s, a significant portion of energy is associated with very low-frequency waves, with the Fourier amplitudes increasing as

frequency decreases. Comparing wave intensities in Fig. 2, example D corresponds to exceptionally low intensity. Therefore, it is unlikely that the energy in the low-frequency range corresponds to forced quadratic nonlinear wave components phase-locked with shorter waves in the spectrum. The kinetic nonlinear theory of shallow-water waves predicts $\omega^{-4/3}$ and ω^{-1} dependencies for the direct and inverse cascades, respectively, in the energy spectrum of surface perturbations [30]. For deep-water conditions, the corresponding power-law dependencies are ω^{-4} and $\omega^{-11/3}$ for the direct and inverse cascades, respectively. The spectrum for case D is plotted in Fig. 3 using logarithmic coordinates and compared with power-law functions. The figure shows that the low-frequency part of the spectrum is well described by the theoretical ω^{-1} dependence for the inverse cascade of nonlinear waves in shallow water. In [8], a similar spectral region, observed in measurements in the shallow Currituck Sound lagoon, was attributed to the contribution of soliton-like long waves.

It is evident that the hydrostatic approximation used in this study is not applicable for discussing the behavior of the high-frequency part of the spectrum. Within the framework of linear dispersive wave theory, the spectral distributions of bottom pressure variations considered here are related to the surface displacement spectra via the transfer function $\cosh^{-2}(kh)$ which does not affect the long-wave region ($kh \ll 1$) but ensures an exponential decay in the high-frequency limit. Indeed, as it can be seen in Fig. 3, the high-frequency part of the spectrum decays much more rapidly than the $\omega^{-4/3}$ and ω^{-4} functions. At the same time, high-frequency dependence follows more closely a power law with a large exponent rather than an exponential decay.

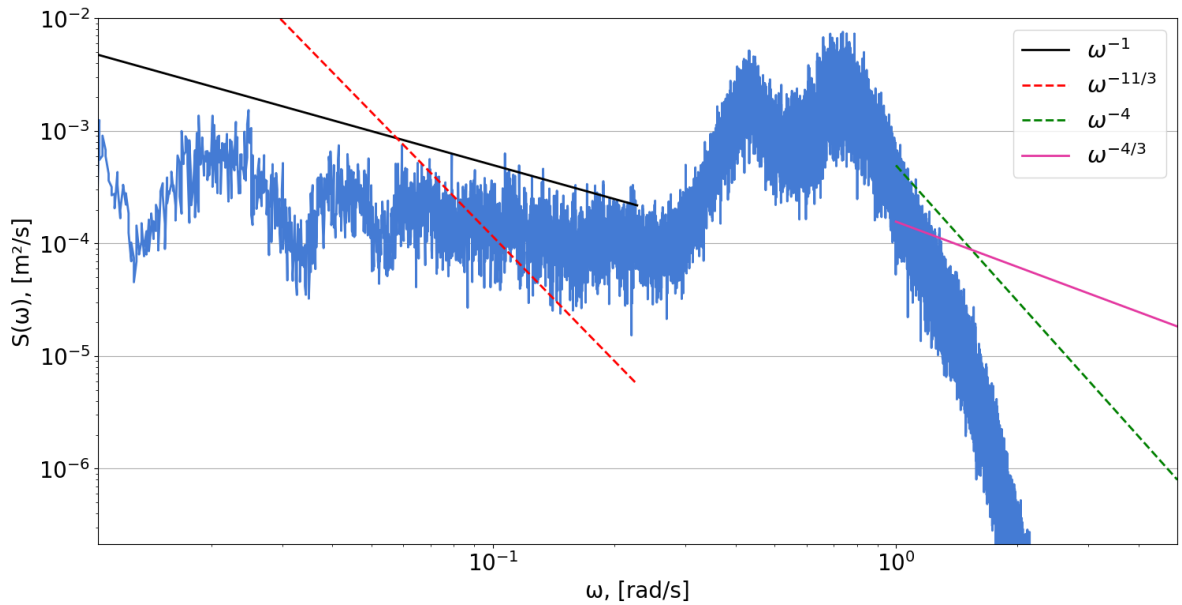


Fig. 3. Energy frequency spectrum for the interval shown in Fig. 2D, plotted in logarithmic coordinates using window averaging. Theoretical slopes of the direct and inverse nonlinear wave cascades are shown with solid and dashed lines for shallow- and deep-water limits, respectively.

This may be related either to a data processing artifact (due to violations of sampling periodicity) or to the presence of natural and instrumental noise. The inaccuracy of reconstructing surface displacement from bottom pressure measurements using a linear transfer function is well known - applying an exponential factor when converting the data, due to the presence of noise in the spectral tail, leads to divergence in the Fourier distributions of the reconstructed surface displacement fields. This issue is not addressed in the present study.

2. EFFECT OF FREQUENCY SPECTRUM WIDTH ON THE PROBABILITY DISTRIBUTION OF WAVE HEIGHTS

The simplest strictly derived distribution of wave heights is given by the Rayleigh law for the probability of exceeding a given wave height:

$$P(H) = \exp\left[-2\left(\frac{H}{H_s}\right)^2\right] \quad (2)$$

for a linear superposition of independent wave harmonics with random phases (we will discuss this model more in detail later in this text) and a relatively narrow spectrum [4], [14]. In this case, the water surface displacement at a point represents a random Gaussian process. In distribution (2), the quantity H denotes the wave height (the vertical distance between the crest maximum and the adjacent trough minimum – positive and negative amplitudes respectively). The reference height H_s (significant wave height) is equal to $4\sigma^2$, or in oceanography it is often defined as the mean of the highest one-third of waves in the sample, $H_s = H_{1/3}$, which provides better agreement with experimental data (with this choice, numerical simulation data also better match distribution (2) [27]).

For a narrow spectrum, $H_{1/3} \approx 4.004\sigma$ and the choice of definition for H_s has little practical effect [14]. However, as the spectral bandwidth increases, $H_{1/3}$ and 4σ begin to diverge. The influence of finite spectral width was analyzed in [13] as a perturbation to the Rayleigh distribution. The resulting wave height distribution retains the form of (2), with the reference height varying according to the spectral bandwidth (see also [14]).

$$H_s = 4\sigma \sqrt{1 - \left(\frac{\pi^2}{8} - \frac{1}{2}\right)v^2} \approx 4\sigma(1 - 0.367v^2). \quad (3)$$

Here, the spectral bandwidth parameter v is defined in terms of the spectral moments m_n of the frequency spectrum $S(\omega)$:

$$\nu^2 = \frac{m_0 m_2}{m_1^2} - 1 = \frac{\int_0^\infty (\omega - \bar{\omega})^2 S(\omega) d\omega}{\bar{\omega}^2 \int_0^\infty S(\omega) d\omega}, \quad \bar{\omega} = \int_0^\infty \omega S(\omega) d\omega, \quad (4)$$

$$m_n = \int_0^\infty \omega^n S(\omega) d\omega. \quad (5)$$

Equation (3) implies that as the spectral bandwidth increases, the effective value of H_s decreases for a given σ , which, according to distribution (2), should lead to a reduction in the probability of high waves. Other theoretical wave height probability distributions that account for finite spectral width can be found, for example, in [16].

Alternative definitions of spectral bandwidth have also been proposed in the literature. Another estimate based on spectral moments arises naturally when constructing probability distributions for the maxima of random waves [7]:

$$\chi^2 = 1 - \frac{m_2^2}{m_0 m_4}, \quad 0 < \chi^2 \leq 1, \quad (6)$$

as well as the so-called Goda parameter Q_p [9] also referred to as the peak enhancement factor:

$$Q_p = \frac{2}{\sigma^4} \int_0^\infty \omega S^2(\omega) d\omega. \quad (7)$$

Unlike the parameters in (4) and (6), high values of the Goda parameter correspond to narrow-spectrum conditions (pronounced spectral peak).

The spectral width χ is defined using the fourth spectral moment, which presents a challenge for real instrumental data with slowly decaying spectra (or, in general, containing a constant high-frequency noise component). In the present study, this characteristic was computed using its theoretical relation to the mean number of local maxima of the surface displacement per half wavelength, n_m :

$$\chi = \frac{2\sqrt{n_m^2 - n_m}}{2n_m - 1}. \quad (8)$$

It is discussed in [16] and has been shown to perform well for the data from our numerical simulations of ensembles of linear waves [29]. We will return to this estimate later and demonstrate its theoretical basis.

The dependence of wave height probability distributions in field data on spectral bandwidth parameters is shown in Fig. 4. To construct these plots, the original data set was divided into four subsets [5, 31] according to the corresponding spectral bandwidth parameter: the two characteristics defined via spectral moments and the Goda parameter. The intervals of spectral width values were selected to obtain subsets with a representative number of waves, corresponding to markedly different spectral parameter values, and to ensure that transitioning from one interval to another produced a sequential adjustment of the probability distribution. As shown in Fig. 4A, dividing the data by different intervals of the parameter ν does not lead to any noticeable systematic change in the probability distribution. A significant difference in the range of very high waves (relative to H_s) is observed for records with a narrow spectrum according to the χ parameter (Fig. 4B); the subset $\chi < 0.4$ corresponds to relatively narrow-spectrum conditions, in which practically all local maxima of the surface displacement represent crest peaks, with $n_m \approx 0.968$. The most consistent and pronounced adjustment of the probability distributions occurs with changes in the Goda parameter (Fig. 4C). As spectral bandwidth increases, the probability of high waves relative to the 4σ level decreases systematically. For waves exceeding the conventional rogue wave threshold, $H/4\sigma > 2$ (a total of 225 such waves in the analyzed data), the occurrence probability in narrow-spectrum conditions $Q_p \geq 1.4$ is several times higher than in broad-spectrum conditions $Q_p < 0.8$. Thus, the behavior of the probability distribution function is most sensitive and consistent with respect to spectral bandwidth expressed through the Goda parameter.

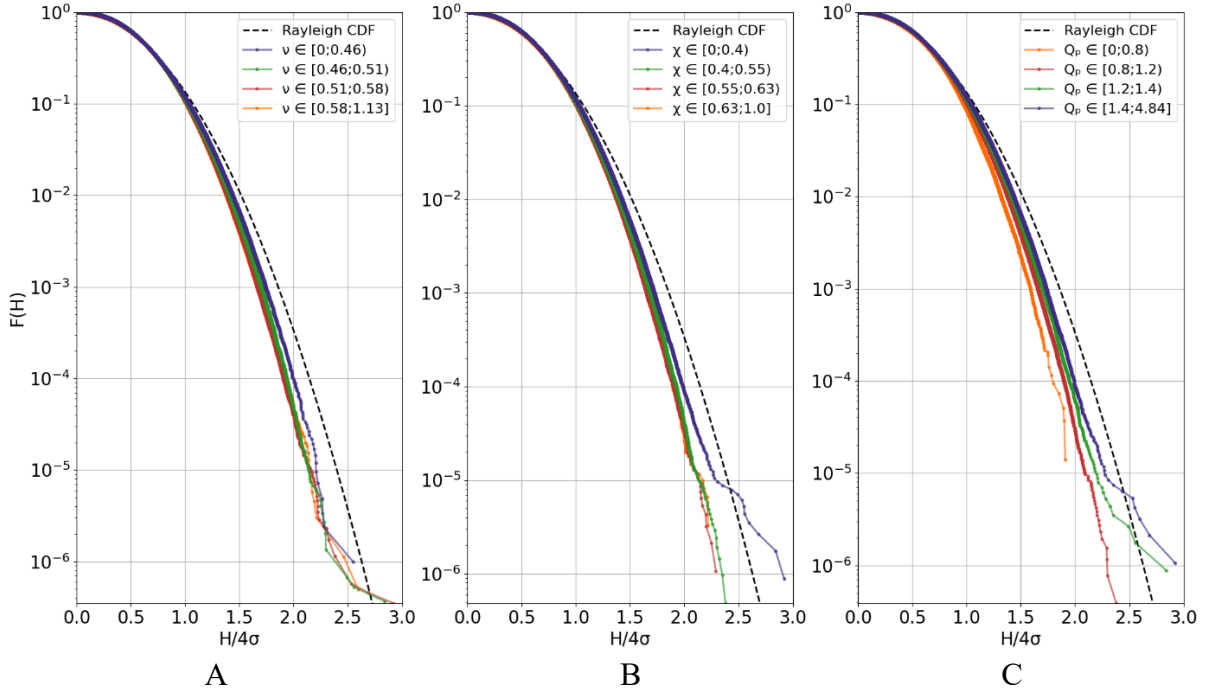


Fig. 4. Wave height probability distributions for different intervals of spectral bandwidth, defined according to the ν (A), χ (B) и Q_p (C). Here and in the following, the reference Rayleigh distribution (2) for $H_s = 4\sigma$ is shown with a dashed line.

The diagram shows the frequency of occurrence of sea state conditions in the plane of the Goda parameter (horizontal axis) and shallow-water nonlinearity $a = \sigma/h$ (vertical axis) is presented in Fig. 5. Here and in the following, the characteristic wavenumber k is determined from the estimated value of the angular frequency $\bar{\omega}$ (4) using the linear dispersion relation for gravity waves on a finite depth h . Figure 5 indicates that the most frequently realized conditions correspond to a Goda parameter value of $Q_p \approx 1$ and weak wave nonlinearity. It can be seen that the region of small nonlinearity parameters, $a < 0.05$, is characterized by a peak in the distribution at somewhat lower Q_p values: at lower nonlinearity, conditions with broader spectra are realized. For conditions with significant nonlinearity, the shape of the contour lines for events with equal occurrence suggests a hyperbolic-type relationship, $aQ_p \propto \text{Const}$, indicating that events with higher nonlinearity are associated with comparatively broader spectra.

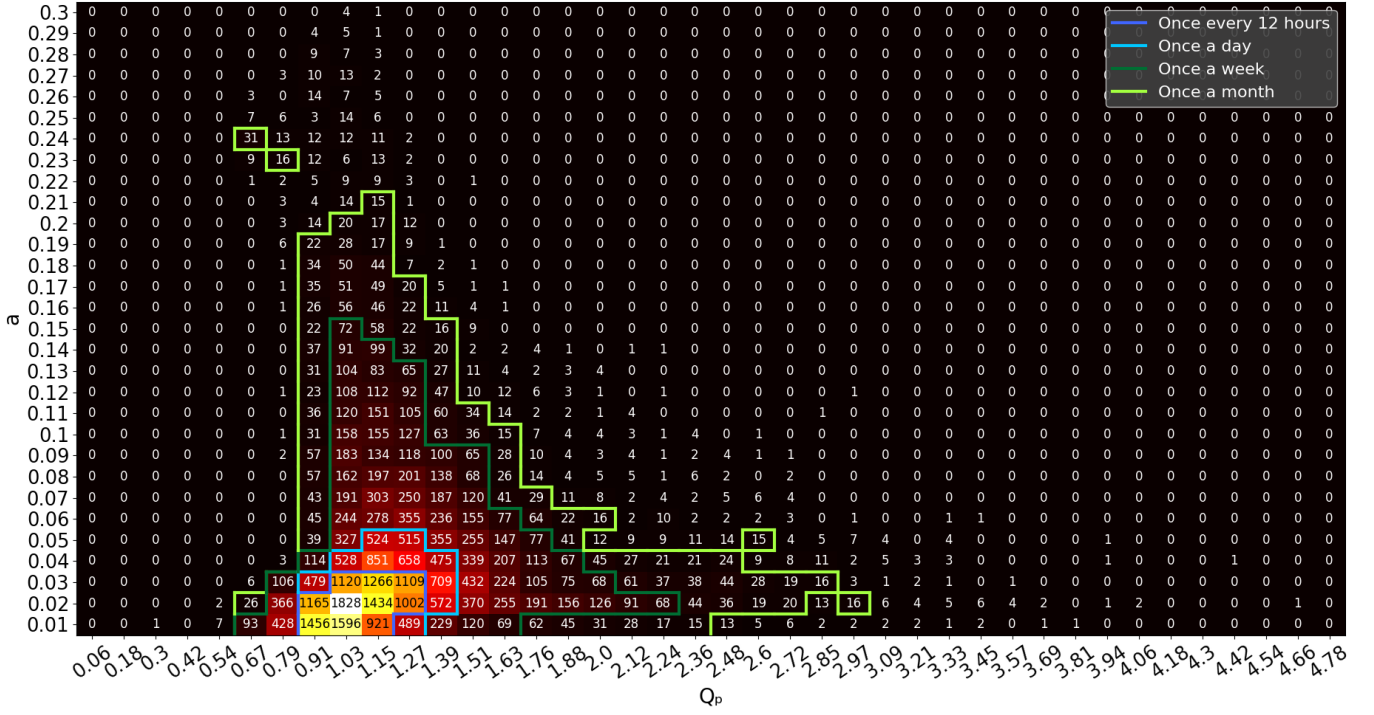


Fig. 5. Frequency distribution of conditions with different spectral bandwidth Q_p and wave nonlinearity a . The numbers in the cells indicate the number of 20-minute intervals. Contours of wave conditions occurring once per month, once per week, once per day, and once every 12 hours are highlighted by color (see line coding in the figure).

This observation raises the question of whether the reduction in the probability of high waves with increasing spectral bandwidth, evident in Fig. 4C, is indeed due to changes in spectral width itself, rather than being a consequence of generally more intense waves associated with broader frequency spectra. This issue is addressed on Fig. 6. In Fig. 6A, distributions analogous to those in Fig. 4C are shown, but only for data with low nonlinearity, $a < 0.04$. In Fig. 6B and 6C, wave height probability distributions are presented for different intervals of the nonlinearity parameter a for two subsets of the Goda parameter: $0.8 \leq Q_p < 1.2$ (Fig. 6B) and $1.2 \leq Q_p < 1.4$ (Fig. 6C). The data in Fig. 6 indicate the presence of two independent dependencies: the probability of high waves decreases with increasing nonlinearity a (as already noted in our previous work [5]), as shown in Fig. 6B, C; it also decreases with increasing spectral width (decreasing Q_p) under conditions of very weak nonlinearity, as shown in Fig. 6A.

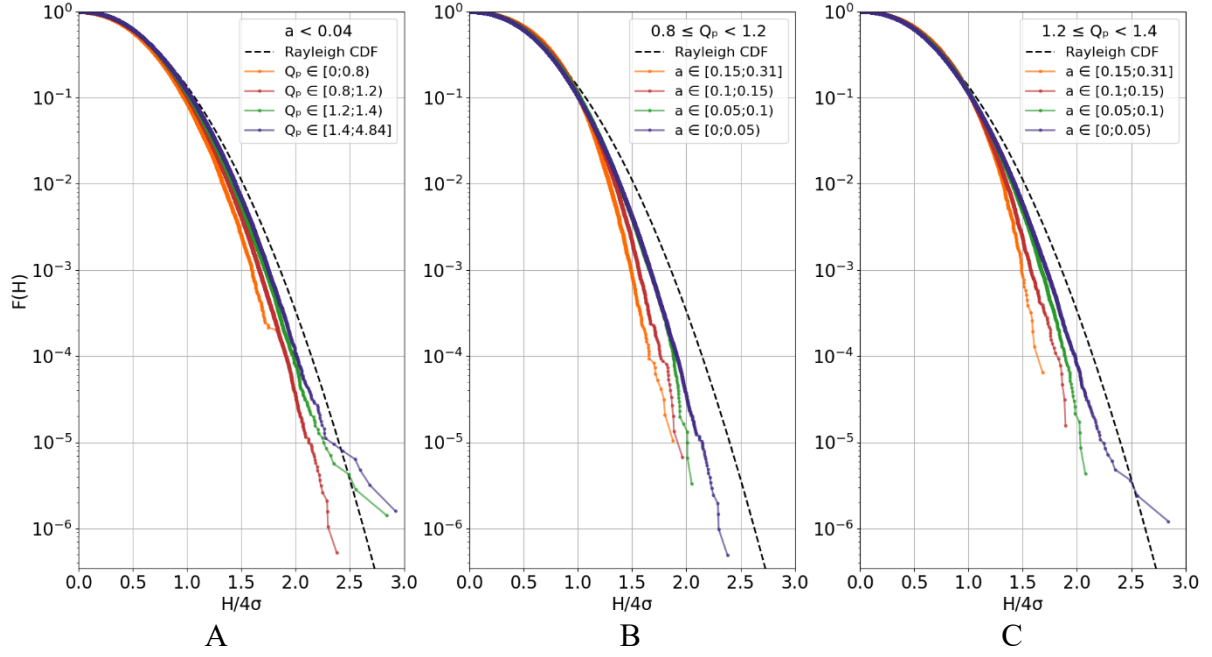


Fig. 6. Wave height probability distributions: for different intervals of spectral bandwidth Q_p at low nonlinearity $a < 0.04$ (A), and for different intervals of the nonlinearity parameter a for the subsets $0.8 \leq Q_p < 1.2$ (B) and $1.2 \leq Q_p < 1.4$ (C).

3. RELATIONSHIP OF COMPOSITE SPECTRAL PARAMETERS (SIMILARITY PARAMETERS) WITH THE PROBABILITY OF OBSERVING HIGH WAVES

The spectral bandwidth estimate is used in the Benjamin–Feir index (BFI), introduced in [11, 17]. It characterizes the ratio of nonlinearity, represented by the wave steepness $k\sigma$, to the strength of dispersive effects, estimated as the relative spectral width of wavenumbers $\delta k = \Delta k/k$,

$$BFI = 2\sqrt{2} \frac{k\sigma}{\delta_k}. \quad (9)$$

It is important to note that this characteristic was introduced for deep-water conditions, where the characteristic spatial scale is the wavelength rather than the water depth. Although the dependence of the probability of extreme wave occurrence on BFI values has been repeatedly demonstrated in numerical and laboratory model experiments [20], its use in field conditions remains controversial, partly due to the very low values of this index observed under realistic conditions, which are significantly below the theoretical threshold for instability, $BFI > 1$ [12, 22].

A modification of the BFI (9) accounting for finite water depth can be expressed in the following form [24]:

$$BFI = \sqrt{2} \frac{k\sigma}{\delta_\omega} \Gamma(kh) = \sqrt{2\pi} k\sigma Q_p \Gamma(kh), \quad (10)$$

Where δ_ω is the relative spectral bandwidth. Following [24], the most robust estimate of the spectral bandwidth δ_ω in (10) is obtained using the Goda parameter Q_p (7). This conclusion is consistent with the results of the analysis presented in the previous section of this study.

The coefficient $\Gamma > 0$ in (10) depends only on the dimensionless depth kh and is plotted in Fig. 7 (the analytical expression can be found in [24]). This parameter accounts for the effects of depth-dependent nonlinear and dispersive coefficients, as well as the relationship between the temporal and spatial spectral widths; in the deep-water limit, $\Gamma \rightarrow 1$. This dependence is already pronounced at relatively large but finite values of $kh < 5$. At $kh \approx 1.36$, the parameter Γ becomes zero (due to the degeneration of four-wave interactions), and at even shallower depths, Γ increases toward infinity, reflecting the emergence of a different natural length scale (the basin depth) and a qualitative change in the pattern of nonlinear wave interactions in shallow water (dominated by 3-wave processes). For $kh < 1.36$ planar modulated waves are stabilized, and the modulation instability criterion based on the *BFI* loses its meaning.

At the same time, in the range $0.5 < kh < 1.36$, so-called oblique wave perturbations (i.e., oriented at an angle to the propagation direction) remain unstable, and the *BFI* continues to characterize the balance between nonlinear and dispersive effects for quasi-sinusoidal waves. As shown in Fig. 7, for $kh < 1$, the coefficient Γ exceeds unity, the value corresponding to deep-water conditions. This may lead to formally high *BFI* values, which, however, do not indicate a high probability of modulation instability. For the three-dimensional problem, [23] noted the existence of a specific modulation angle relative to the wave direction, for which the degeneracy of the dispersion coefficient may enhance the role of wave self-modulation. It should be noted, however, that these considerations are based on the nonlinear Schrödinger equation (first-order nonlinear-dispersive theory), whose applicability is limited to quasi-sinusoidal waves - a condition already violated in shallow water at relatively small wave amplitudes. Several generalizations of the modulation instability indicator incorporating directional wave spectrum information have been proposed in the literature (see, e.g., [12, 15]). A mechanism of nonlinear wave self-modulation in intersecting wave systems at certain angles has also been discussed [18]. However, in the absence of directional spectrum data, these theories cannot be applied.

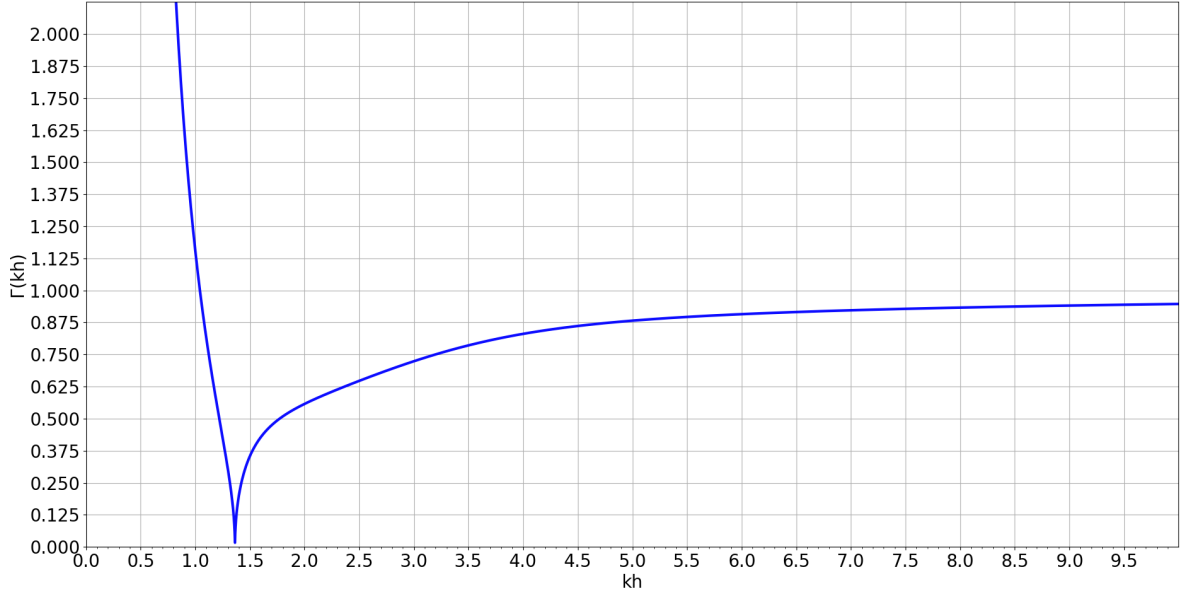


Fig. 7. Dependence of the Goda parameter on the dimensionless depth kh .

In shallow water ($kh \sim 1$) the balance between quadratic nonlinearity and wave dispersion should be characterized by the well-known Ursell parameter Ur :

$$Ur = \frac{3}{2} \frac{k\sigma}{(kh)^3} = \frac{3}{2} \frac{a}{(kh)^2}. \quad (11)$$

In this case, the spectral width can be interpreted as a measure of deviation from the long-wave limit, so that $\Delta k \approx k$. For isolated nonlinear waves (solitons), the Ursell parameter satisfies $Ur = O(1)$ regardless of the wave amplitude. (The coefficient in definition (11) is chosen in accordance with the nonlinear parameter μ for waves of arbitrary depth from [28]. In the deep-water limit, $\mu(kh \rightarrow \infty) \rightarrow k\sigma$, while in shallow water, $\mu(kh \rightarrow 0) \rightarrow Ur$).

The influence of the Ursell parameter on wave height probability distributions is examined in Fig. 8. As before, the wavenumber k in (11) is determined from the linear dispersion relation for waves of given frequency $\bar{\omega}$ on water of constant depth. In Fig. 8A, distributions are shown for several non-overlapping intervals of these dimensionless similarity parameters. As follows from (9) – (11), in addition to spectral width, the governing dimensionless parameters here are wave steepness $k\sigma$ and depth kh . Figures 8B and 8C show distributions for data corresponding to different intervals of the components forming the Ursell parameter: characteristic steepness and dimensionless depth, respectively. Comparison of the panels in Fig. 8 indicates that wave steepness has a significant effect on the probability distribution (Fig. 8B); however, the most systematic separation of the curves for different parameter intervals occurs in Fig. 8A. According to Fig. 8A, the probability of high waves decreases monotonically with increasing

Ur ; the maximum Ur value in the analyzed records is approximately 3.7. In Fig. 8C, only the distribution corresponding to the largest depths, $kh > 1.4$, differs noticeably from the others, while the remaining distributions diverge only in the tails, where the amount of data is small.

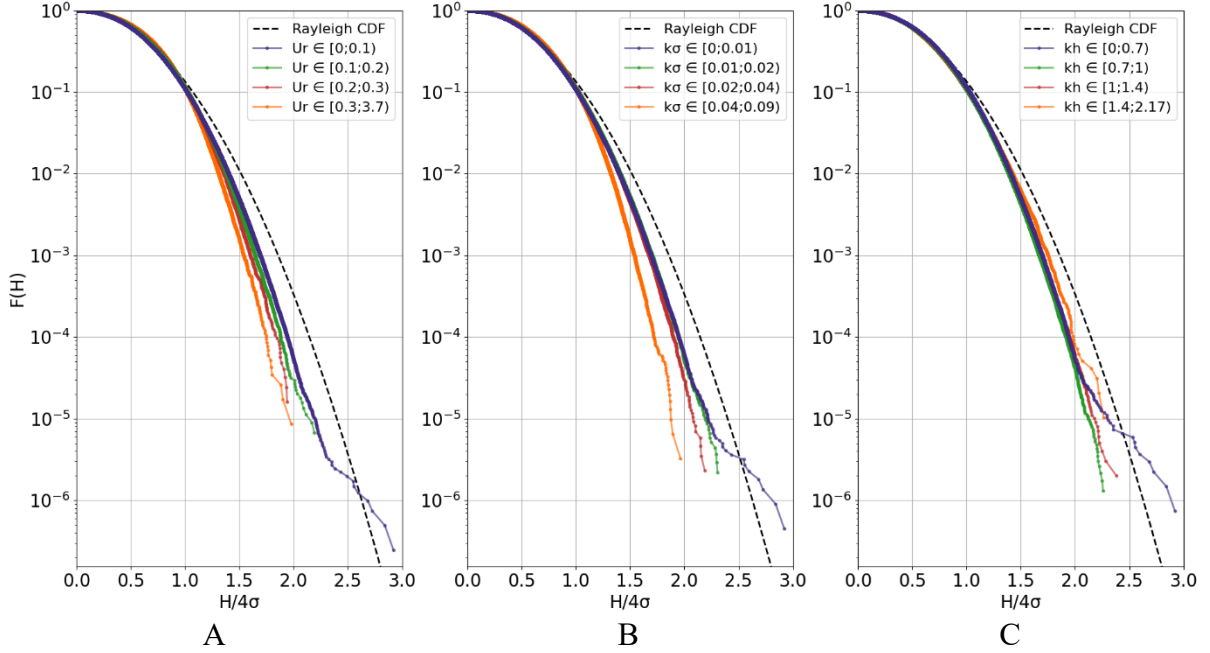


Fig. 8. Wave heights probability distributions for different intervals of Ur (A), $k\sigma$ (B) and kh (C).

Wave height probability distributions for different intervals of the BFI parameter (10) are presented in Fig. 9. In Fig. 9A, the curves are constructed using the deep-water limit of the BFI , denoted BFI_∞ , where the factor Γ in (10) is set to unity, $BFI_\infty = \sqrt{2}k\sigma/\delta_\omega$. In Fig. 9B, the depth-dependent factor $\Gamma(kh)$ is taken into account. The maximum value of BFI_∞ for the analyzed data is approximately 0.3. The corresponding maximum of the BFI index accounting for finite depth exceeds 2.6, but this occurs due to large values of $\Gamma > 1$ in shallow water, where the applicability of BFI is questionable. From Fig. 9A, a trend of decreasing probability of high waves with increasing BFI_∞ , is observed, which is opposite to the original purpose of this parameter as an indicator of modulational instability leading to large-amplitude waves. When the depth-dependent factor Γ is included (Fig. 9B), the distributions behave qualitatively similarly to Fig. 9A, but their dependence on BFI weakens; only the subset with the highest $BFI > 0.4$ values, shows a significant deviation from the rest and is characterized by a markedly lower probability of extreme waves. This behavior is consistent with the interpretation of the BFI in shallow water as an index characterizing the degree of nonlinear longitudinal demodulation of waves, for which theoretical predictions indicate a reduction in the probability of high waves [19] within the framework of weakly nonlinear theory.

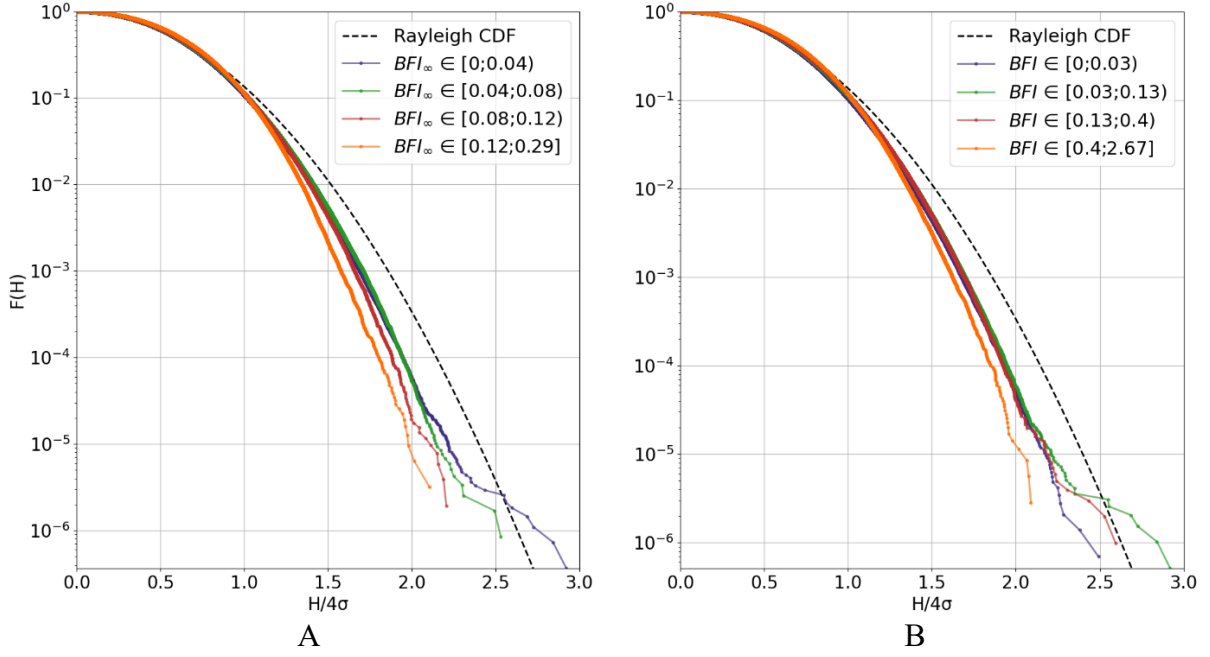


Fig. 9. Wave height probability distributions for different intervals of the BFI parameter (10): in the deep-water limit BFI_∞ (with $\Gamma = 1$ in formula (10)) (A) and accounting for the depth-dependent factor $\Gamma(kh)$ (B).

4. DEPENDENCE OF PROBABILITY DISTRIBUTIONS OF INDIVIDUAL WAVE AMPLITUDES ON SPECTRAL WIDTH

In this chapter, we turn to numerical modeling to examine how spectral width, expressed in various forms, affects the behavior of probability distribution functions of dimensionless wave amplitudes for model data. Wave fields are simulated using the following expression [4, 7, 16]:

$$\eta(x, t) = \sum_{i=1}^N a_i \cos(\omega_i t - k_i x + \varphi_i) \quad (12)$$

Here, $\eta(x, t)$ denotes the water surface displacement, $a_i = \sqrt{2S(\omega)\Delta\omega}$ are the spectral amplitudes, $S(\omega)$ is the wave energy spectrum, and $\Delta\omega$ is the spectral discretization interval, $\Delta\omega = 2\pi/L$, where L is the duration of the realization, $L = N\Delta t$, $\Delta t = 2\pi/\Omega$, N is the number of harmonics, $\Omega = \max(\omega_i)$, $\omega_i = i\Delta\omega$. The wavenumbers k_i are determined from the dispersion relation. The fine-tuning of the model parameters to optimize the generation of its realizations was discussed previously in my bachelor's thesis, as well as in [29].

We fix the spatial coordinate at zero:

$$\eta(x = 0, t) = \sum_{i=1}^N a_i \cos(\omega_i t + \varphi_i), \quad (13)$$

Note that all the following considerations also apply to the case $\eta(x, t = \text{const})$. In the resulting wave field (i.e., in a realization of formula (13)), we identify individual waves, compute their amplitudes A , and then normalize these amplitudes by the parameter a_s , defined as the mean of the highest one-third of amplitudes in the realization. The Rayleigh distribution for the amplitudes, in terms of the normalized amplitude $a = A/a_s$, is then expressed as:

$$F_R(a) = e^{-2a^2} \quad (14)$$

We now construct the probability distribution function of the dimensionless amplitudes of individual waves. In [29], it was noted that this function coincides with the Rayleigh distribution (14) in the case of a narrow spectrum $S(\omega)$, whereas significant deviations - particularly for small waves - are observed in the case of a broad spectrum. In that study, spectral width was characterized using parameter (4).

To investigate the influence of the spectrum on amplitude distribution functions, we attempt to define a model spectrum $S(\omega)$ ourselves. The spectrum must be chosen to be sufficiently flexible in terms of its achievable width (as measured by both spectral width formulas, (4) and (6)), so that its parameters allow for both extremely narrow and extremely broad spectra. For this purpose, a parabolic spectrum (Fig. 10) is well suited, defined differently on the left and right sides of the peak, with the left branch fixed at the origin.

$$S_p(\omega) = \begin{cases} -K_L(\omega - \omega_0)^2 + Q, & \omega \in (0; \omega_0] \\ -K_R(\omega - \omega_0)^2 + Q, & \omega \in (\omega_0; \omega_0 + \omega_x] \\ 0, & \omega \in (\omega_0 + \omega_x; +\infty) \end{cases} \quad (15)$$

Here, K_L and K_R are parameters controlling the width of the left and right branches, respectively. The parameter $Q = \sqrt[3]{K_L K_R / (\sqrt{K_L} + \sqrt{K_R})^2}$ ensures the conservation of total energy (the area under the curve remains constant regardless of the values of K_L and K_R), with $\omega_0 = \sqrt{Q/K_L}$, and $\omega_x = \sqrt{Q/K_R}$.

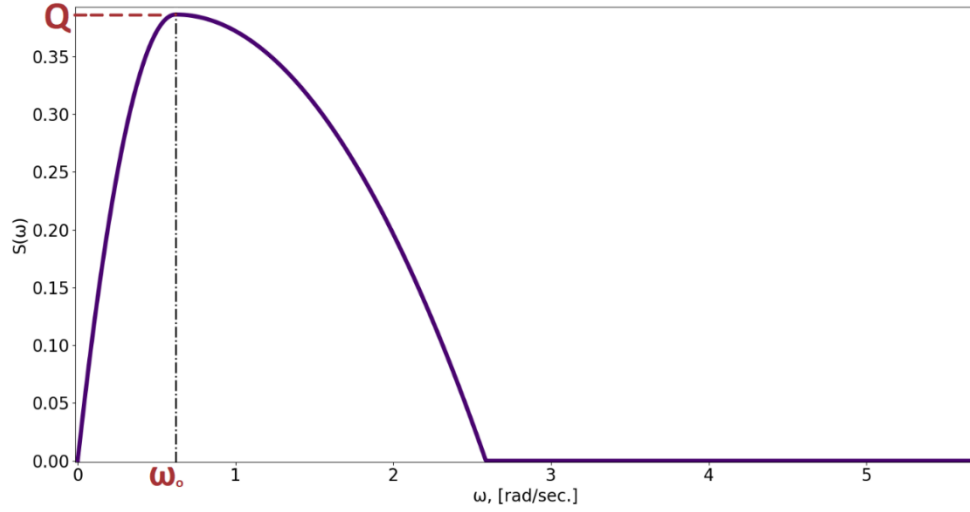
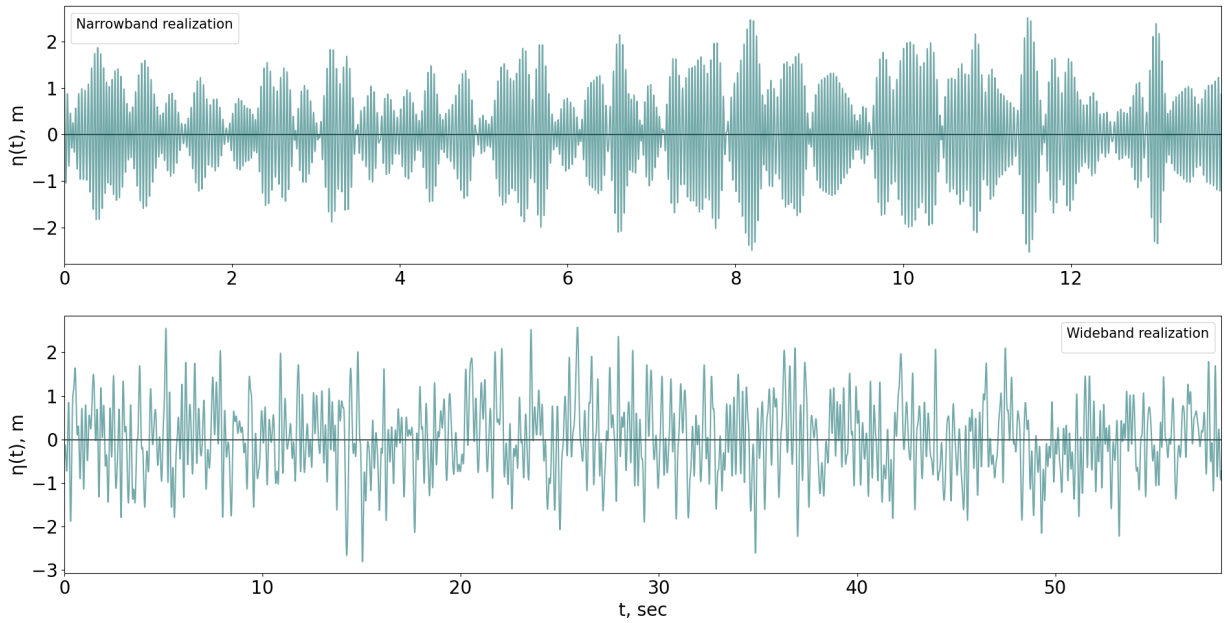


Fig. 10. Plot of the parabolic spectrum showing different branch widths.

Let us now, for definiteness, define the width of the resulting spectrum using the parameter ν , as given in formula (4). We simulate several wave fields (Fig. 11) and, based on them, construct in Fig. 12 the amplitude probability distribution functions for three cases of spectral width (relatively narrow, relatively broad, and intermediate) to examine how the relative positions of these curves change with respect to the Rayleigh distribution (14).



ν .

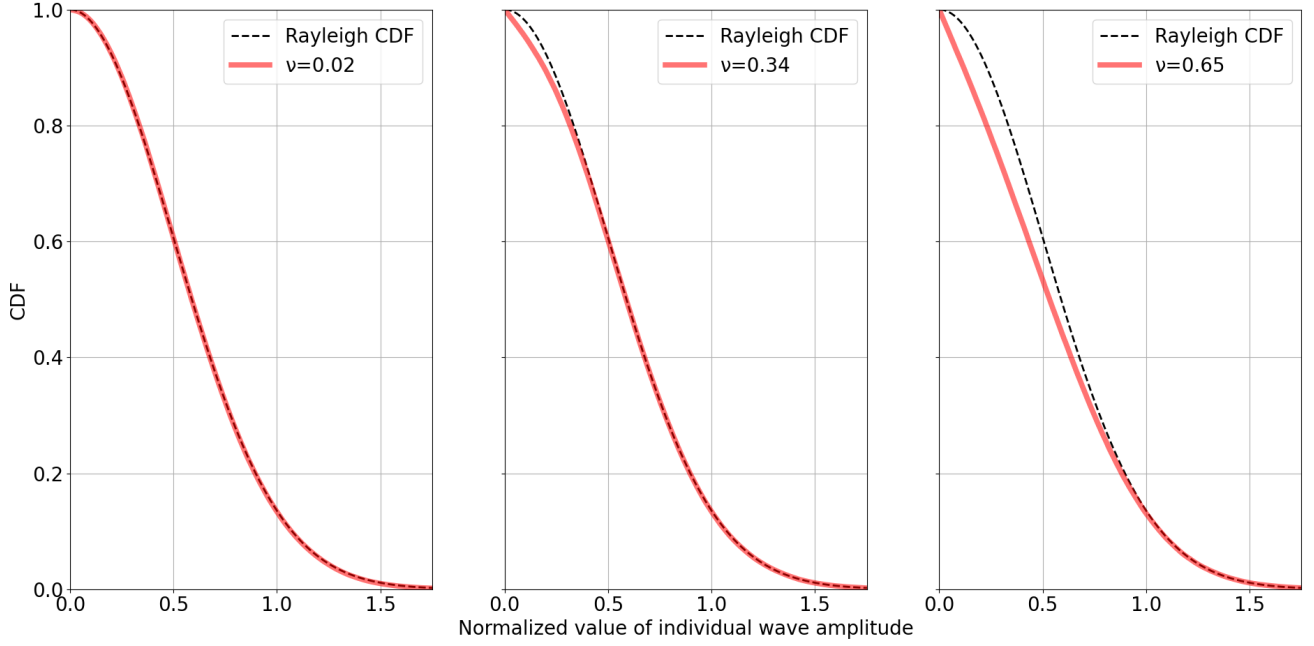


Fig. 12. Comparison with the Rayleigh distribution of the probability functions for positive amplitudes of individual waves generated using spectra with different values of ν .

It should be noted that the Rayleigh distribution itself directly follows from (13) under the assumption of a narrow spectrum (a formal derivation can be found, for example, in [14]). However, a formula for the amplitude distribution function in the case of a spectrum of arbitrary width remains unknown. An attempt to derive it analytically was made in [7], where the main result is the F_M law for the probability distribution of the local maxima of function (13). We rewrite it here in a form convenient for our purposes, using normalized amplitudes:

$$F_M(a) = \frac{1 + e^{-2a^2} \sqrt{1 - \chi^2} - \operatorname{erf} \left[\frac{a\sqrt{2}}{\chi} \right] + e^{-2a^2} \sqrt{1 - \chi^2} \operatorname{erf} \left[\frac{a\sqrt{2}\sqrt{1 - \chi^2}}{\chi} \right]}{1 + \sqrt{1 - \chi^2}} \quad (16)$$

Fig. 13 illustrates, through a numerical experiment, the validity of distribution (16): practical and theoretical probability functions for the local maxima are constructed using the same range of spectral width values, demonstrating that the curves indeed coincide.

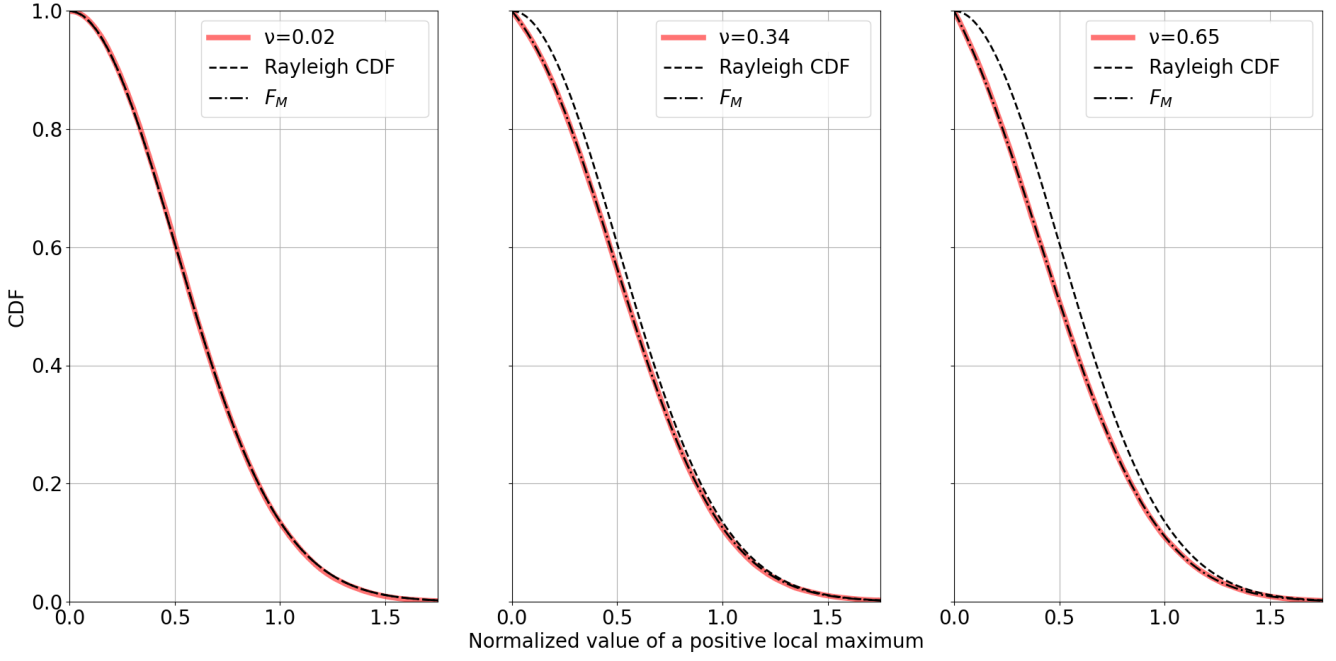


Fig. 13. Comparison of the distribution function of positive local maxima of the function with the $F_M(a)$ distribution and the Rayleigh distribution for different spectral widths.

However, in the general case, the distribution law for the local maxima of the function does not coincide with the amplitude distribution law (denoted here for convenience as F_A). This arises from the fact that the amplitude of an individual wave, which we define based on zero crossings, is taken as the largest of all local maxima within that wave. In turn, in the case of a broad spectrum, an individual wave may contain multiple local maxima. This is clearly illustrated by the example in Fig. 14.

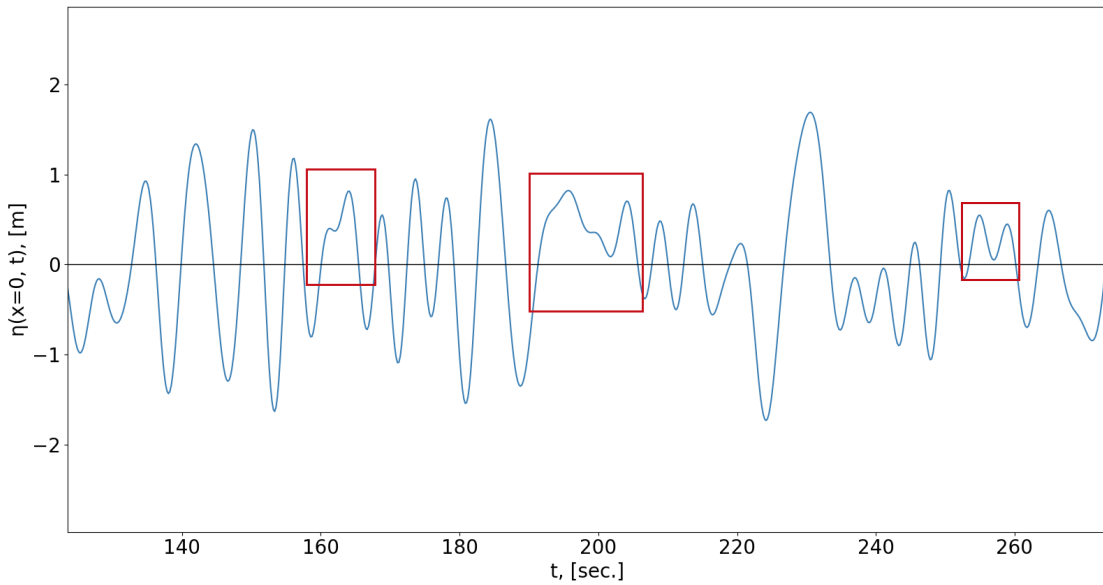


Fig. 14. Individual waves containing multiple local maxima.

The distribution F_A can be estimated through the quantity $n_A(a)$ – the number of amplitudes greater than or equal to a , and N_A – the total number of amplitudes, as follows:

$$F_A(a) = \frac{n_A(a)}{N_A} \quad (17)$$

Then, the distribution F_M can be written as:

$$F_M(a) = \frac{n_M(a)}{N_M} = \frac{n_A(a) + n_L(a)}{N_A + N_L} \quad (18)$$

Here, $n_M(a) = n_A(a) + n_L(a)$ represents the number of local maxima with values greater than or equal to a , and $N_M = N_A + N_L$ is the total number of local maxima. The quantities $n_L(a)$ and N_L denote, respectively, the number of local maxima that are not absolute within an individual wave and whose values are greater than or equal to a and the total number of such maxima. We introduce a new distribution of “non-global” maxima, $F_L(a) = n_L(a)/N_L$. From (18), it is straightforward to see that the distribution $F_M(a)$ decomposes into F_A and F_L with some coefficients:

$$F_M(a) = \frac{N_A}{N_M} F_A(a) + \frac{N_L}{N_M} F_L(a) = \frac{N_A}{N_M} F_A(a) + \left(1 - \frac{N_A}{N_M}\right) F_L(a) \quad (19)$$

We can now rewrite the last equality (19), taking into account that we have an analytical expression (16), which we slightly reformulate by representing it as the sum of two functions with certain coefficients:

$$\frac{N_A}{N_M} F_A + \left(1 - \frac{N_A}{N_M}\right) F_L = \frac{\sqrt{1 - \chi^2}}{1 + \sqrt{1 - \chi^2}} e^{-2a^2} \left(1 + \operatorname{erf}\left[\frac{a\sqrt{2}\sqrt{1 - \chi^2}}{\chi}\right]\right) + \frac{1}{1 + \sqrt{1 - \chi^2}} \left(1 - \operatorname{erf}\left[\frac{a\sqrt{2}}{\chi}\right]\right) \quad (20)$$

Written in this form, the equality at first glance appears to uniquely determine both the distributions F_A and F_L , as well as the ratio N_A/N_M , especially given that $\sqrt{1 - \chi^2}/(1 + \sqrt{1 - \chi^2}) = 1 - 1/(1 + \sqrt{1 - \chi^2})$. However, upon closer examination, it turns out that the expressions in the parentheses do not even satisfy the definition of a distribution function.

The parameter $N_A/N_M = 1 - N_L/N_M$ is indeed of considerable interest, as it essentially represents the ratio of the number of local maxima that are global within an individual wave to the total number of local maxima. Let us denote this ratio by $\rho = N_A/N_M$. Clearly, $0 < \rho \leq 1$; a value of zero is inadmissible since an individual wave in formula (2) cannot lack absolute maxima altogether. The relationship between this ratio and the spectral width parameter χ was derived analytically and has already been mentioned above in equation (8). We now rewrite this relationship in a form more closely aligned with our current discussion:

$$\rho(\chi) = \frac{2\sqrt{1-\chi^2}}{1+\sqrt{1-\chi^2}} \quad (21)$$

$$\chi(\rho) = \frac{2\sqrt{1-\rho}}{2-\rho} \quad (22)$$

It should be noted that, in fact, these relationships can be “intuitively inferred” even without knowing the analytical form from expression (20): on both the left- and right-hand sides we have ρ and some expression involving χ , with the constraints $\rho(\chi = 0) = 1$ and $\rho(\chi = 1) = 0$. By straightforwardly selecting functions that satisfy these conditions and are contained, in one form or another, in the coefficients on the right-hand side of expression (20), one can compare them with the curve obtained from numerical simulations. It should also be noted that the chosen spectrum (15) is sufficiently flexible to allow modeling across nearly the entire range of the spectral width parameter, $0 \leq \chi \leq 1$. The resulting simulation, which confirms equality (21) and the subsequent equality (22), is presented in Fig. 15.

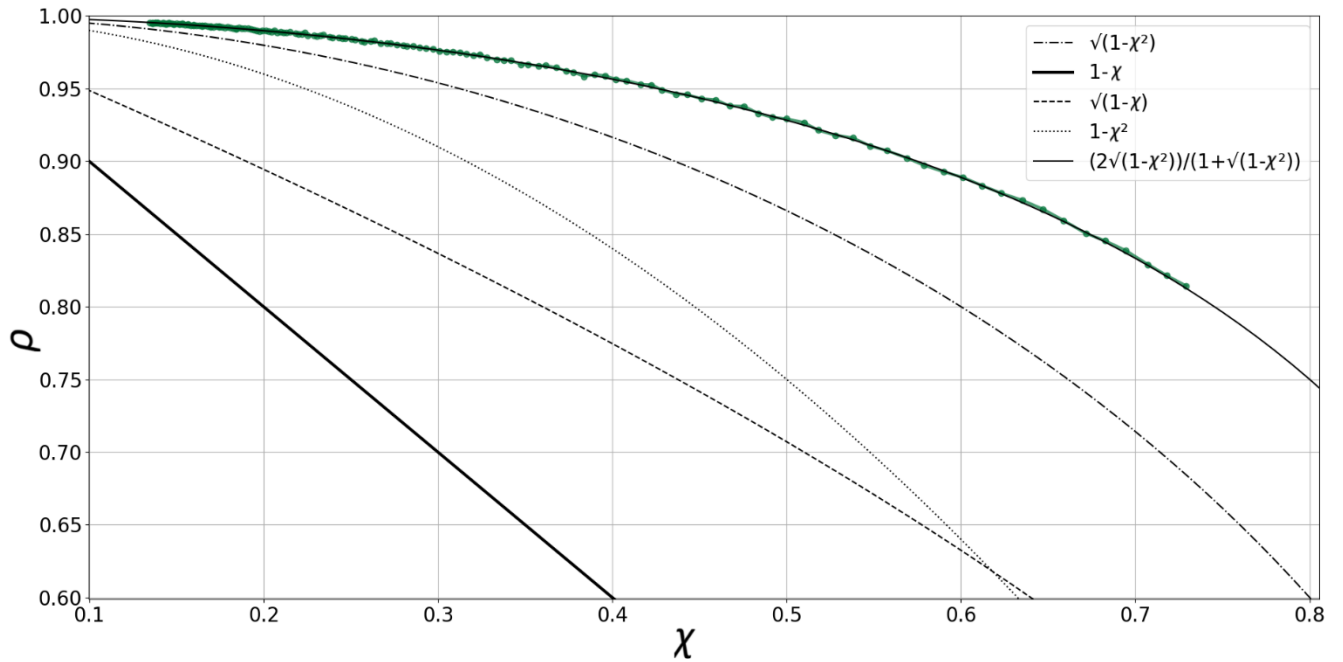


Fig. 15. Dependence of the ratio of global maxima to the total number of maxima on χ .

Finally, equation (20) can be rewritten taking into account (19) and (21):

$$F_M = \rho F_A + (1 - \rho) F_L = \rho \frac{e^{-2a^2}}{2} \left(1 + \operatorname{erf} \left[\frac{a\sqrt{2}\sqrt{1-\chi^2}}{\chi} \right] \right) + (1 - \rho) \frac{1 - \operatorname{erf} \left[\frac{a\sqrt{2}}{\chi} \right]}{1 - \sqrt{1-\chi^2}} \quad (23)$$

Let us introduce the following notations:

$$F_1(a) = \frac{e^{-2a^2}}{2} \left(1 + \operatorname{erf} \left[\frac{a\sqrt{2}\sqrt{1-\chi^2}}{\chi} \right] \right); F_2(a) = \frac{1 - \operatorname{erf} \left[\frac{a\sqrt{2}}{\chi} \right]}{1 - \sqrt{1-\chi^2}} \quad (24)$$

As already noted, these functions are not true distribution functions: $F_1(0) = 1/2$, while $F_2(0) = 1/(1 - \sqrt{1-\chi^2})$. Figure 16 shows the graphs of these functions (in the limit $\chi \rightarrow 1$):

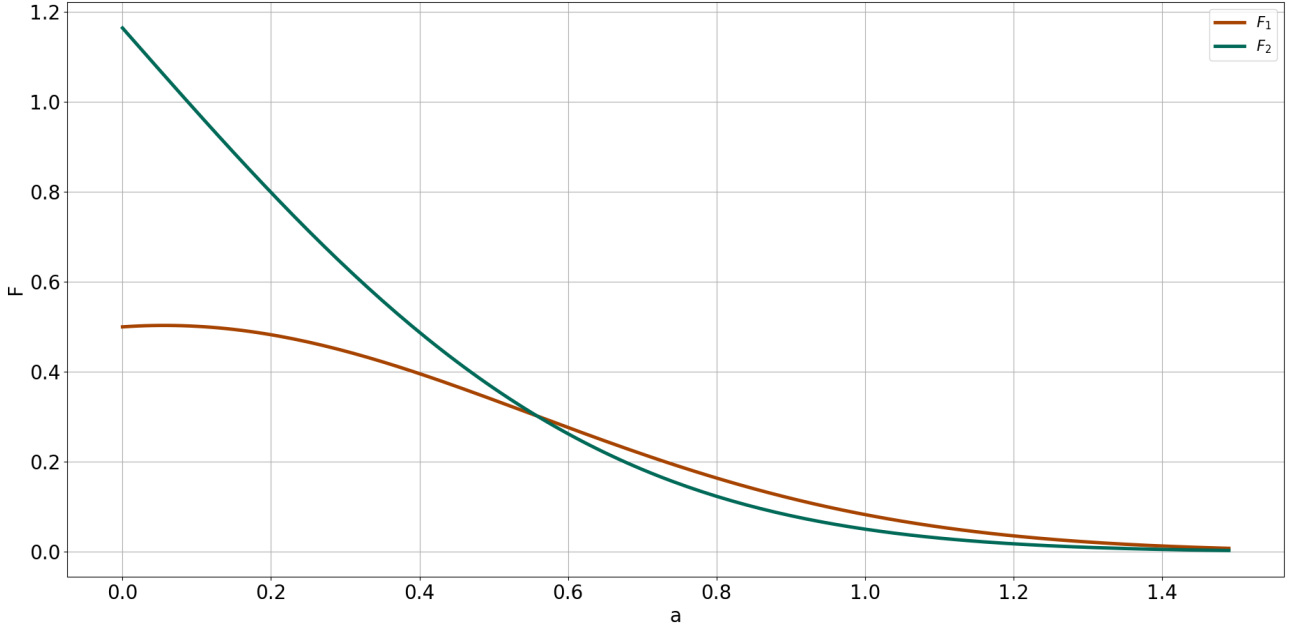


Fig. 16. Graphs of the functions F_1 and F_2 .

Suppose there exists a function $\gamma(a)$ such that $F_1(a) + \gamma(a) = F_A(a)$. Then, to preserve the equality $F_M = \rho F_A + (1 - \rho)F_L$, this addition must be subtracted from $F_2(a)$, i.e.:

$$F_M = \frac{2\sqrt{1-\chi^2}}{1 + \sqrt{1-\chi^2}} (F_1 + \gamma(a)) + \frac{1 - \sqrt{1-\chi^2}}{1 + \sqrt{1-\chi^2}} \left(F_2 - \frac{2\sqrt{1-\chi^2}}{1 - \sqrt{1-\chi^2}} \gamma(a) \right) \quad (25)$$

Thus,

$$F_L(a) = F_2 - \frac{2\sqrt{1-\chi^2}}{1 - \sqrt{1-\chi^2}} \gamma(a) \quad (26)$$

Obviously, to satisfy $F_L(0) = F_A(0) = 1$, it is necessary to require $\gamma(0) = 1/2$. More stringent conditions follow from the normalization constraints:

$$\int_0^\infty (1 - [F_1 + \gamma])'_a da = 1 \quad \text{and} \quad \int_0^\infty \left(1 - \left[F_2 - \frac{2\sqrt{1-\chi^2}}{1 - \sqrt{1-\chi^2}} \gamma \right] \right)'_a da = 1, \quad (27)$$

since we are seeking a function that is itself a proper probability distribution. We then attempt to determine the form of the additional function γ numerically: again, for different values of χ we construct (Fig. 17) the amplitude probability distributions F_1 and $F_A - F_1$:

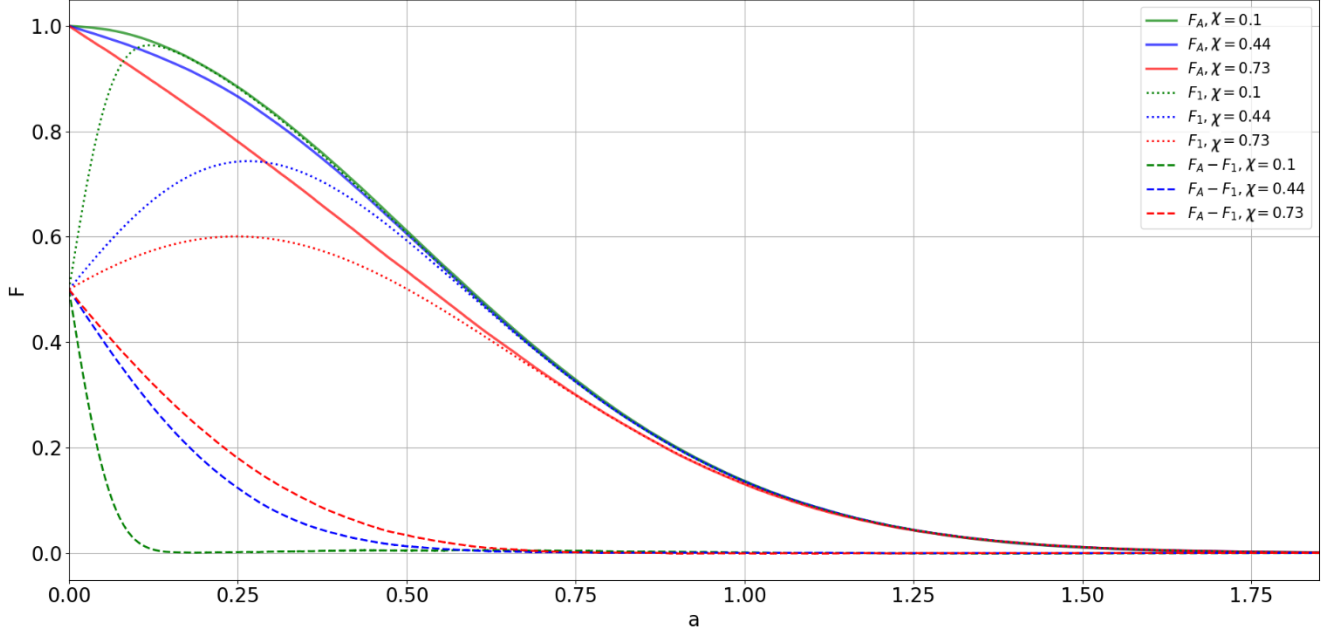


Fig. 17. Numerical amplitude probability distributions and their differences with F_1 .

Thus, based on the shape of the resulting differences, we assume that:

$$\gamma(a) = \frac{1}{2}(1 - \text{erf}[\psi a]) \quad (28)$$

Here, ψ is a parameter controlling the stretching of the function $\gamma(a)$ along the a -axis. It is natural to assume that $\psi = \psi(\chi)$. Let us attempt to derive some restrictions on the function $\psi(\chi)$, for this purpose we check normalization condition (we need F_A to be a cumulative distribution function):

$$\int_0^\infty (1 - [F_1 + \gamma])'_a da = \int_0^\infty \frac{\psi e^{-a^2 \psi^2}}{\sqrt{\pi}} da + 4 \int_0^\infty a F_1 da - \int_0^\infty \frac{e^{-\frac{2a^2}{\chi^2}} \sqrt{2} \sqrt{1 - \chi^2}}{\sqrt{\pi} \chi} da = \frac{\psi}{2\sqrt{\psi^2}} + \frac{1}{2} \quad (29)$$

Let us focus on the first term:

$$\int_0^\infty \frac{\psi e^{-a^2 \psi^2}}{\sqrt{\pi}} da = \frac{\psi}{2\sqrt{\psi^2}} \quad (30)$$

For the sum of integrals in (29) to equal unity, it is necessary that (30) equals 1/2; this condition is satisfied for every $\psi(\chi) > 0$. Therefore, the normalization requirement is fulfilled for all positively

defined functions $\psi(\chi)$ over the interval $\chi \in (0, 1)$. It is easy to see that the same calculations applies to F_L .

So, $\gamma(a)$ indeed completes F_1 and F_2 into proper distribution functions. In Fig. 18, we compare the numerically obtained curves $F_A - F_1$ with the $\gamma(a)$ functions, manually adjusting the value of $\psi(\chi)$ for each χ :

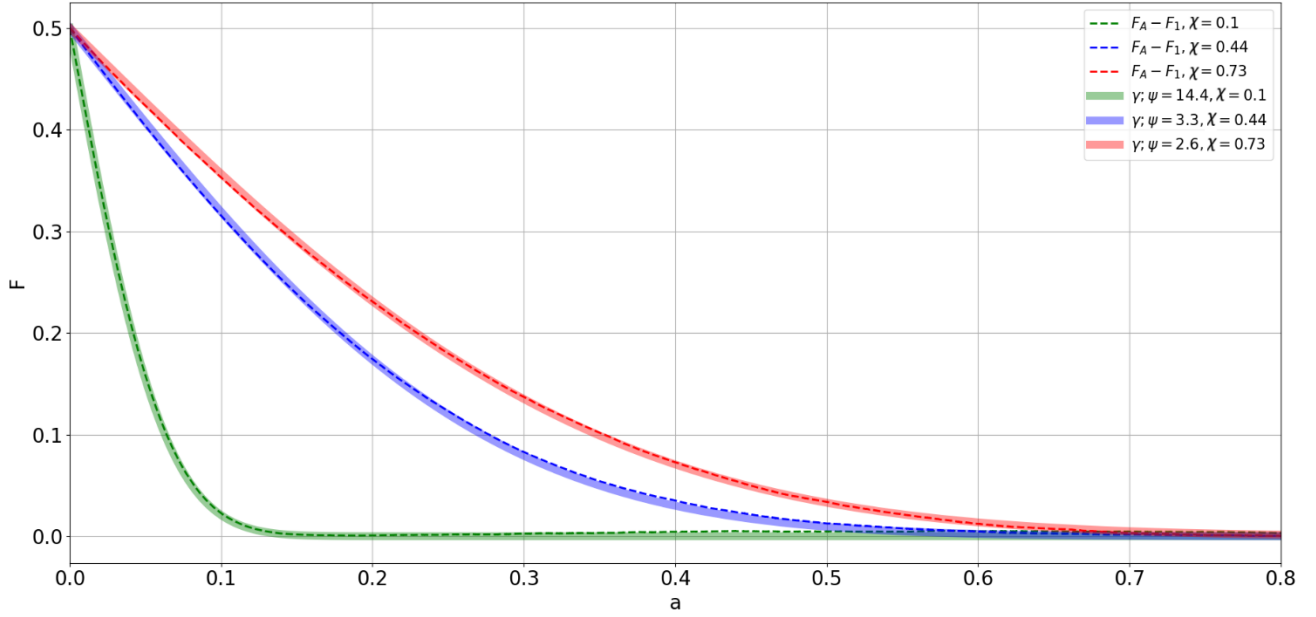


Fig. 18. Comparison of the numerically obtained complement function with the assumed $\gamma(a)$.

Figure 19 shows the numerically obtained graph of $\psi(\chi)$. This curve is well approximated by the following function:

$$\psi^*(\chi) = \frac{\sqrt{2}(1 + \chi^4)}{\chi} \quad (31)$$

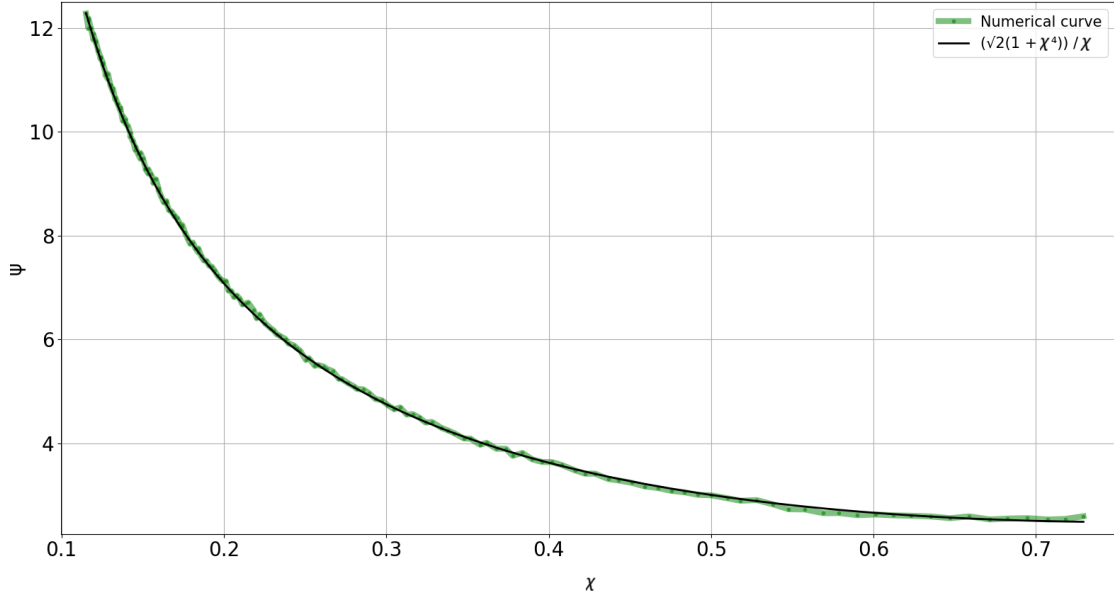


Fig. 19. Numerical approximation of $\psi(\chi)$.

As a result, we assume that the amplitude distribution function can be written as:

$$F_A = \frac{1}{2} \left(1 - \operatorname{erf} \left[\frac{\sqrt{2}(1 + \chi^4)}{\chi} a \right] + e^{-2a^2} \left(1 + \operatorname{erf} \left[\frac{\sqrt{2}\sqrt{1 - \chi^2}}{\chi} a \right] \right) \right) \quad (32)$$

and the distribution function of local maxima that are not global within an individual wave is:

$$F_L = \frac{1 - \operatorname{erf} \left[\frac{\sqrt{2}}{\chi} a \right] - \sqrt{1 - \chi^2} \left(1 - \operatorname{erf} \left[\frac{\sqrt{2}(1 + \chi^4)}{\chi} a \right] \right)}{1 - \sqrt{1 - \chi^2}} \quad (33)$$

Conclusions

In this study, for the first time, an analysis of long-term wind-wave spectra near the coast of Sakhalin Island was conducted using the measurement database of Specialized Design Bureau of the Sakhalin Branch of the Far Eastern Branch of the Russian Academy of Sciences. A high variability of the spectra is demonstrated, including examples of single and multiple wave systems, the development of local wind waves, and the arrival of waves from distant generation areas. Examples are provided of spectra realizations with profiles close to the JONSWAP model spectrum, as well as the occurrence of inverse cascades of nonlinear waves corresponding to shallow-water conditions.

For the measurement site, the occurrence frequency of wave conditions was determined in terms of the spectral width and wave nonlinearity, taking into account the variability of the dimensionless depth parameter due to tides and changes in the wave period. Particular attention was given to the dependence of observed probability distributions of wave heights on the spectral characteristics of the waves. It was shown that the probability of extreme waves decreases with increasing spectral width, as well as with an increase in the ratio of characteristic wave amplitude to water depth. The clearest and most consistent reorganization of probability distribution is observed when spectral width is assessed via the Goda parameter. A systematic reduction in the probability of high waves is also demonstrated for the Ursell parameter, which characterizes the ratio of characteristic shallow-water wave nonlinearity to dispersive effects.

The Benjamin–Feir index (BFI), introduced in the literature as an indicator of increased probability of extreme waves, has extremely limited applicability for the analyzed dataset due to the shallow depth at the measurement site. It is proportional to the ratio of wave steepness to spectral width, so the effects of nonlinearity and spectral width on wave probability discussed above tend to compensate each other in this parameter. Under shallow-water conditions, the effect of nonlinear wave self-modulation changes to demodulation, and the phenomenon becomes highly dependent on directional spreading of the waves, which cannot be captured by a single instrument at a fixed point. Nevertheless, the field data show a certain dependence on the modulation instability parameter, but of an opposite nature: as BFI increases, the probability of extreme waves decreases.

The experimentally obtained probability distributions lie below the reference Rayleigh distribution, and in the range of extreme waves they decrease further with increasing nonlinearity or spectral width. For this reason, the probability of so-called rogue waves, defined by $H > 8\sigma$, is lower (or much lower) than the value 3.35×10^{-4} predicted by the Rayleigh distribution. At the same time, in the present study the reconstruction of surface displacement fields was based on hydrostatic theory, whose

applicability to some of the measurements is questionable and may alter the probabilistic characteristics of extreme waves. In future work, we plan to employ more accurate reconstruction procedures.

It should be noted that, due to the relatively limited size of the dataset, cases with different numbers of spectral peaks were analyzed together. For this reason, the “broad-spectrum” conditions considered in the present study effectively include situations with the coexistence of multiple wave systems, which may exhibit specific wave statistics.

In the concluding part of this study, we proposed (without a rigorous proof) a formula, which consistent with the numerical simulation results, which defines the probability distribution of individual wave amplitudes for a spectrum of arbitrary width. We also demonstrated why and how spectral width can be estimated from observational data without directly computing the spectrum (i.e., without transforming into Fourier space). However, the details and limits of applicability of this approach remain to be established in future works.

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